Univalence rules out uniform double-negation elimination for all types

The Univalent Foundations Program

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Editorial note. Univalence forces a uniform operation on types to respect the Boolean swap; dependent transport of the double-negation argument then creates a forbidden fixed point. The original path-transport calculation is retained.

Allowed standard prerequisites (editorial).

- Basic univalent type theory. Identity elimination, function extensionality, univalence, dependent sums/products, propositional truncation, standard path transport and the explicitly supplied higher-inductive elimination rules. Standard elementary real/ordinal facts may be used; all target-specific local construction proofs are retained.

Source reference. Original reference 2.9.4: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1324–1324. Original book navigation or standard prerequisite reference; target construction and its local proof chain are retained in body spans.

Source reference. Original reference 2.12.6: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1600–1600. Original book navigation or standard prerequisite reference; target construction and its local proof chain are retained in body spans.

Source reference. Original reference 2.10: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1631–1631. Original book navigation or standard prerequisite reference; target construction and its local proof chain are retained in body spans.

Source reference. Original reference 1.11: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1689–1689. Original book navigation or standard prerequisite reference; target construction and its local proof chain are retained in body spans.

Source reference. Original reference 2.15.7: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1909–1909. Original book navigation or standard prerequisite reference; target construction and its local proof chain are retained in body spans.

Source reference. Original reference 3.1.9: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1911–1911. Original book navigation or standard prerequisite reference; target construction and its local proof chain are retained in body spans.

3.1 Propositions as types?

Until now, we have been following the straightforward "propositions as types" philosophy described in original reference 1.11, according to which English phrases such as "there exists an $x : A$ such that $P(x)$" are interpreted by corresponding types such as $\sum_{(x:A)} P(x)$, with the proof of a statement being regarded as judging some specific element to inhabit that type. However, we have also seen some ways in which the "logic" resulting from this reading seems unfamiliar to a classical mathematician. For instance, in original reference 2.15.7 we saw that the statement

> "If for all $x : X$ there exists an $a : A(x)$ such that $P(x, a)$, then there exists a function $g : \prod_{(x:X)} A(x)$ such that for all $x : X$ we have $P(x, g(x))$", (3.1.1)

which looks like the classical *axiom of choice*, is always true under this reading. This is a noteworthy, and often useful, feature of the propositions-as-types logic, but it also illustrates how significantly it differs from the classical interpretation of logic, under which the axiom of choice is not a logical truth, but an additional "axiom".

On the other hand, we can now also show that corresponding statements looking like the classical *law of double negation* and *law of excluded middle* are incompatible with the univalence axiom.

Theorem 3.1.2. *It is not the case that for all $A : \mathcal{U}$ we have $\neg(\neg A) \to A$.*

Proof. Recall that $\neg A \equiv (A \to \mathbf{0})$. We also read "it is not the case that …" as the operator $\neg$. Thus, in order to prove this statement, it suffices to assume given some $f : \prod_{(A:\mathcal{U})}(\neg\neg A \to A)$ and construct an element of $\mathbf{0}$.

The idea of the following proof is to observe that f, like any function in type theory, is "continuous". By univalence, this implies that f is *natural* with respect to equivalences of types. From this, and a fixed-point-free autoequivalence, we will be able to extract a contradiction.

Let $e : \mathbf{2} \simeq \mathbf{2}$ be the equivalence defined by $e(1_{\mathbf{2}}) :\equiv 0_{\mathbf{2}}$ and $e(0_{\mathbf{2}}) :\equiv 1_{\mathbf{2}}$, as in original reference 3.1.9. Let $p : \mathbf{2} = \mathbf{2}$ be the path corresponding to e by univalence, i.e. $p :\equiv \mathsf{ua}(e)$. Then we have $f(\mathbf{2}) : \neg\neg\mathbf{2} \to \mathbf{2}$ and

$$\mathsf{apd}_f(p) : \mathsf{transport}^{A\mapsto(\neg\neg A\to A)}(p, f(\mathbf{2})) = f(\mathbf{2}).$$

Hence, for any $u : \neg\neg\mathbf{2}$, we have

$$\mathsf{happly}(\mathsf{apd}_f(p), u) : \mathsf{transport}^{A\mapsto(\neg\neg A\to A)}(p, f(\mathbf{2}))(u) = f(\mathbf{2})(u).$$

Now by (2.9.4), transporting $f(\mathbf{2}) : \neg\neg\mathbf{2} \to \mathbf{2}$ along p in the type family $A \mapsto (\neg\neg A \to A)$ is equal to the function which transports its argument along p^{-1} in the type family $A \mapsto \neg\neg A$, applies $f(\mathbf{2})$, then transports the result along p in the type family $A \mapsto A$:

$$\mathsf{transport}^{A\mapsto(\neg\neg A\to A)}(p, f(\mathbf{2}))(u) = \mathsf{transport}^{A\mapsto A}(p, f(\mathbf{2})(\mathsf{transport}^{A\mapsto\neg\neg A}(p^{-1}, u))).$$

However, any two points $u, v : \neg\neg\mathbf{2}$ are equal by function extensionality, since for any $x : \neg\mathbf{2}$ we have $u(x) : \mathbf{0}$ and thus we can derive any conclusion, in particular $u(x) = v(x)$. Thus, we have $\mathsf{transport}^{A\mapsto\neg\neg A}(p^{-1}, u) = u$, and so from $\mathsf{happly}(\mathsf{apd}_f(p), u)$ we obtain an equality

$$\mathsf{transport}^{A\mapsto A}(p, f(\mathbf{2})(u)) = f(\mathbf{2})(u).$$

Finally, as discussed in original reference 2.10, transporting in the type family $A \mapsto A$ along the path $p \equiv \mathsf{ua}(e)$ is equivalent to applying the equivalence e; thus we have

$$e(f(\mathbf{2})(u)) = f(\mathbf{2})(u). \tag{3.1.3}$$

However, we can also prove that

$$\prod_{x:\mathbf{2}} \neg(e(x) = x). \tag{3.1.4}$$

This follows from a case analysis on x: both cases are immediate from the definition of e and the fact that $0_{\mathbf{2}} \neq 1_{\mathbf{2}}$ (original reference 2.12.6). Thus, applying (3.1.4) to $f(\mathbf{2})(u)$ and (3.1.3), we obtain an element of $\mathbf{0}$. □